

Intermediate Code Generation (Compiler Design) - Complete Interview Guide

Phase of the compiler: Intermediate code generation comes after syntax analysis and semantic analysis. The compiler converts the checked program into an internal form called an **intermediate representation (IR)**. This IR is easier to optimize, analyze, and translate into machine code than raw source code.

This guide covers the most important intermediate-code-generation topics for SDE placements, online assessments, university viva questions, and technical interviews.

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Big Picture: Where Intermediate Code Generation Fits

```
Source Program
  |
  v
Lexical Analysis      -> tokens
  |
  v
Syntax Analysis       -> parse tree / AST
  |
  v
Semantic Analysis     -> typed / annotated AST
  |
  v
Intermediate Code Gen -> IR, TAC, CFG, SSA
  |
  v
Optimization          -> improved IR
  |
```

V
Code Generation -> assembly / machine code / bytecode

One-line interview answer: Intermediate code generation converts a semantically valid program into a machine-independent internal form so the compiler can optimize it and later generate target-specific code.

1. Intermediate Representation

1. Overview

Definition: An **intermediate representation (IR)** is an internal form of a program used by a compiler between the source language and the final target code.

Instead of directly translating C, Java, Python, or Rust source code into machine code, most compilers translate source code into IR first. The IR is then analyzed, optimized, and converted into assembly, bytecode, or machine code.

Why it matters:

- It separates the compiler front end from the back end.
- It makes optimization easier.
- It allows one compiler infrastructure to support many languages and many processors.
- It gives the compiler a simpler representation than source code but a more flexible representation than machine code.

Where it is used in real systems:

- LLVM uses LLVM IR.
- GCC uses GIMPLE and RTL.
- Java compilers emit JVM bytecode.
- .NET compilers emit CIL.
- JavaScript engines use multiple internal IRs for JIT optimization.
- Databases use query plans, which behave like IR for SQL.

Why interviewers ask about it: IR is the center of modern compiler design. If you understand IR, you can explain optimization, code generation, SSA, CFGs, register allocation, and JIT compilation more clearly.

2. Core Idea

The main idea is to convert a complex source program into a simpler internal program.

Intuition: Source code is written for humans. Machine code is written for processors. IR is written for the compiler.

Real-world analogy: Think of IR like an architectural blueprint. A customer may describe a house in natural language, and workers eventually build it using concrete and bricks. The blueprint sits in the middle: precise enough for engineers, but not tied to one worker's tool.

Small example:

```
x = a + b * c;
```

Possible three-address IR:

```
t1 = b * c
t2 = a + t1
x  = t2
```

The IR breaks a complex expression into small operations that are easier to optimize and translate.

Step-by-step:

- 1. Parser builds a syntax structure for `x = a + b * c`.
- 2. Semantic analysis verifies declarations and types.
- 3. Intermediate code generation emits simple operations.
- 4. Optimizer may simplify or reorder operations.
- 5. Back end maps operations to target instructions.

3. Important Subtopics

Subtopic	Meaning	Why It Matters	Example	Common Interview Angle
High-level IR	Close to source language	Keeps loops, objects, and function calls easy to analyze	AST, typed AST	"Why not generate machine code directly?"
Low-level IR	Close to machine operations	Easier for instruction selection and register allocation	LLVM IR, RTL	"How is IR different from assembly?"
Linear IR	Sequence of instructions	Simple to emit and scan	TAC instructions	"Convert expression to TAC."
Graph IR	Nodes and edges represent data/control dependencies	Good for optimization	CFG, DAG, SSA graph	"Why use a CFG?"
Typed IR	IR values carry types	Prevents invalid transformations	<code>add i32 %a, %b</code>	"Why keep type info after semantic analysis?"
Machine-independent IR	Avoids target-specific assumptions	Enables portability	Same IR to x86 and ARM	"How do compilers support many architectures?"

High-level IR: Keeps source-level concepts like loops, arrays, exceptions, or method calls. It is useful for early analysis.

Low-level IR: Uses simpler instructions closer to machine operations. It is useful for late optimization and code generation.

Linear IR: Stores instructions in a list. It is easy to generate and execute in order.

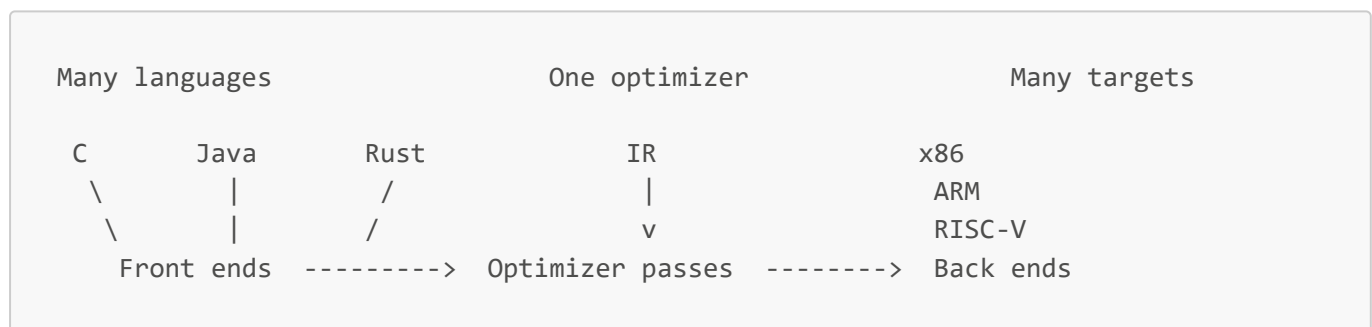
Graph IR: Stores relationships explicitly. It is powerful for global optimization.

4. Real-World Example

Browser JavaScript engine: A browser parses JavaScript into an AST, lowers it into bytecode or IR, profiles hot functions, then JIT-compile selected code into optimized machine code. The IR lets the engine reason about operations like property access, numeric arithmetic, branches, and function calls before producing CPU instructions.

Backend server: A Java backend compiles `.java` files into JVM bytecode. The JVM later interprets or JIT-compile that bytecode. JVM bytecode acts as an intermediate representation between Java source and machine code.

5. Diagrams / Mental Models



Mental model: IR is the compiler's "common currency." Different source languages are converted into the same kind of internal money, then spent by different machine-code generators.

6. Common Interview Questions

Question	Clear Answer	Expected Key Points	Common Mistakes
What is an intermediate representation?	An internal program form used between source code and target code.	Machine independent, optimizable, compiler internal	Calling it only "assembly code"
Why do compilers use IR?	To decouple front end and back end and make optimization easier.	Retargetability, reuse, simpler analysis	Saying "only for speed"
Is bytecode an IR?	Yes, bytecode can be considered an intermediate representation,	JVM bytecode, CIL, portability	Thinking IR must be invisible to users

Question	Clear Answer	Expected Key Points	Common Mistakes
	especially in VM-based languages.		
How is IR different from source code?	IR is compiler-oriented, simpler, explicit, and often normalized.	Temporaries, simple operations, explicit control flow	Saying source and IR are same syntax
How is IR different from machine code?	IR is usually machine independent; machine code is target-specific binary instructions.	Portability, abstraction level	Saying IR always runs directly on CPU
What are common IR forms?	AST, TAC, bytecode, CFG, SSA, DAGs.	Multiple levels of IR	Naming only TAC
Why not generate machine code directly from AST?	It couples source language details to target hardware and makes optimization harder.	Separation of concerns, reuse	Ignoring optimization
What is a good IR property?	It should be easy to generate, analyze, optimize, and translate.	Simplicity plus expressiveness	Saying "as detailed as source"
Can one compiler have multiple IRs?	Yes. Modern compilers often lower through several IR levels.	High-level to low-level pipeline	Assuming exactly one IR
Where does IR generation happen?	After parsing and semantic analysis, before optimization and code generation.	Compiler middle end	Placing it before syntax analysis

7. Deep-Dive Questions

Question	Answer
Why do modern compilers use multiple IR levels?	Different optimizations need different abstraction levels. High-level IR is good for language-level optimizations, while low-level IR is good for register allocation and instruction selection.
What does "lowering" mean in an IR pipeline?	Lowering converts a high-level representation into a simpler, lower-level representation. For example, a for loop may become labels and conditional jumps.
How does IR help retargetability?	If every source language front end emits the same IR, every target back end only needs to consume that IR. This avoids writing one complete compiler for every language-target pair.
What is the trade-off in IR design?	A high-level IR is easier to relate to source code, while a low-level IR is easier to map to hardware. Good compilers often use both.

Question	Answer
Why is type information useful in IR?	It prevents invalid transformations and helps optimizations. For example, integer addition and floating-point addition have different rules.

8. Comparison Tables

Source Code vs IR vs Machine Code

Feature	Source Code	Intermediate Representation	Machine Code
Main audience	Human programmers	Compiler	CPU
Portability	High	Usually high	Low
Optimization-friendly	Medium	High	Low to medium
Readability	High	Medium	Very low
Hardware-specific	No	Usually no	Yes
Example	<code>x = a + b</code>	<code>t1 = a + b</code>	Binary instructions

High-Level IR vs Low-Level IR

Feature	High-Level IR	Low-Level IR
Close to	Source language	Machine operations
Keeps loops/objects?	Often yes	Usually lowered
Best for	Early analysis	Late code generation
Example	AST, typed AST	LLVM-like operations, RTL
Interview trap	Thinking high-level IR is useless	Thinking low-level IR is machine code

9. Common Mistakes

- Thinking IR is always three-address code.
- Saying IR is the final output of a compiler.
- Confusing IR with parse tree.
- Forgetting that compilers can have multiple IRs.
- Assuming IR must be human-readable.
- Ignoring the role of IR in optimization.
- Saying IR is always machine-independent; some late IRs are target-aware.

10. Edge Cases / Special Cases

- Some simple compilers directly generate code from the AST.
- JIT compilers may generate IR at runtime.
- Bytecode can be stored on disk and executed by a VM.
- Debug information must connect optimized IR back to source lines.

- Optimizations can make IR very different from the original source.

11. How to Explain in Interview

"An intermediate representation is the compiler's internal version of the program. After parsing and semantic checks, the compiler converts source code into IR because IR is easier to analyze, optimize, and translate into different target machines. Examples include AST, three-address code, bytecode, CFGs, and SSA."

12. Quick Revision Notes

- IR sits between source code and target code.
- Main benefits: optimization, portability, modular compiler design.
- Examples: AST, TAC, bytecode, LLVM IR, SSA.
- High-level IR is close to source; low-level IR is close to machine code.
- Interview trap: do not call every IR "assembly."

13. Practice Tasks

1. Convert $x = (a + b) * (c - d)$ into a simple IR.
2. Explain why a compiler for 3 languages and 4 CPUs benefits from a common IR.
3. Compare JVM bytecode and LLVM IR.
4. Draw a compiler pipeline showing where IR appears.
5. Identify whether AST, TAC, CFG, and SSA are linear or graph-like representations.

14. Final Cheat Sheet

Item	Notes
Core definition	Internal program representation between source and target code
Why it matters	Enables optimization, portability, and clean compiler architecture
Most asked questions	What is IR? Why use IR? IR vs machine code?
Common comparisons	Source vs IR vs machine code, high-level vs low-level IR
One-line answer	IR is the compiler-friendly form of a program used for analysis, optimization, and code generation.

2. Abstract Syntax Tree

1. Overview

Definition: An **Abstract Syntax Tree (AST)** is a tree representation of the meaningful syntactic structure of a program.

It is "abstract" because it removes unnecessary grammar details such as punctuation nodes, parentheses used only for grouping, and intermediate grammar productions.

Why it matters:

- It is the main output of parsing in many compilers.
- It captures program structure in a form easier to analyze than raw tokens.
- Semantic analysis, type checking, code generation, formatting tools, linters, and refactoring tools commonly operate on ASTs.

Where it is used in real systems:

- Compilers parse source files into ASTs.
- IDEs use ASTs for autocomplete, rename refactoring, and error highlighting.
- Linters like ESLint inspect JavaScript ASTs.
- Transpilers like Babel transform ASTs.
- Static analyzers inspect ASTs to find bugs or security issues.

Why interviewers ask about it: AST is one of the first practical data structures in compiler design. Interviewers use it to check whether you understand parsing output and how source code becomes a structured object.

2. Core Idea

The AST keeps only the structure needed to understand the program.

Intuition: Tokens are a flat list. An AST adds hierarchy.

Real-world analogy: A sentence diagram in grammar class shows how words relate. An AST does the same for code.

Small example:

```
x = a + b * c;
```

AST:



Step-by-step:

1. Lexer produces tokens: `id(x)`, `=`, `id(a)`, `+`, `id(b)`, `*`, `id(c)`, `;`.

- 2. Parser applies grammar rules.
- 3. AST removes unnecessary punctuation.
- 4. Operator precedence is reflected in the tree: `b * c` is deeper than `a + ...`.
- 5. Later phases traverse the tree.

3. Important Subtopics

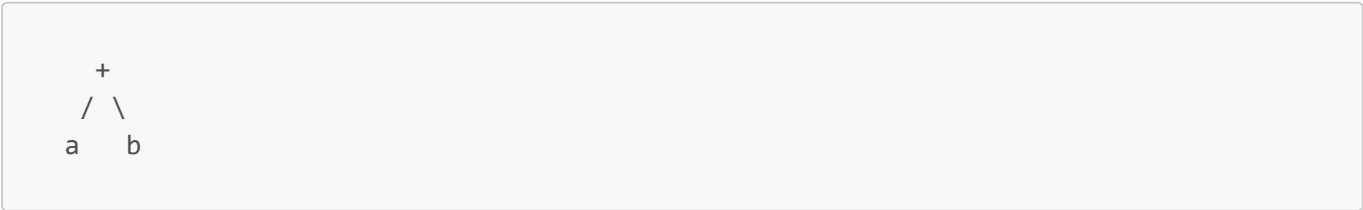
Subtopic	What It Means	Why It Matters	Example	Common Interview Angle
Parse tree vs AST	Parse tree includes all grammar symbols; AST keeps essential structure	AST is smaller and more useful	Parentheses may disappear	"Difference between parse tree and AST?"
Expression nodes	Nodes for operators and operands	Represents computation	<code>+</code> , <code>*</code> , variable nodes	"Draw AST for expression."
Statement nodes	Nodes for assignments, loops, conditionals	Represents program actions	<code>If</code> , <code>While</code> , <code>Return</code>	"How is control flow represented?"
Declaration nodes	Nodes for variables, functions, classes	Used in symbol-table construction	<code>FunctionDecl</code> , <code>VarDecl</code>	"Where are declarations stored?"
Annotated AST	AST plus semantic info	Useful for type checking and IR generation	node type = <code>int</code>	"What happens after parsing?"
AST traversal	Walking the tree recursively or iteratively	Used by visitors, analyzers, compilers	pre-order traversal	"How would you generate code from AST?"

Parse tree vs AST example:

Expression:

```
(a + b)
```

Parse tree may include nonterminals like `Expr`, `Term`, `Factor`, and punctuation. AST usually keeps only:



4. Real-World Example

IDE rename refactoring: When you rename a variable in an IDE, the tool should not blindly replace every matching string. It parses the code into an AST, resolves symbols, and renames only the correct variable

references.

Example:

```
int count = 0;
printf("count");
```

The string "count" should not be renamed as a variable. AST plus semantic info prevents that mistake.

5. Diagrams / Mental Models



AST Node Table

Source Construct	Possible AST Node
x = y + 1	Assign(name=x, value=BinaryOp(+))
if (x > 0)	If(condition=Compare(>))
while (i < n)	While(condition=Compare(<))
return x	Return(value=x)
int f(int a)	FunctionDecl(name=f, params=[a])

6. Common Interview Questions

Question	Clear Answer	Expected Key Points	Common Mistakes
What is an AST?	A tree showing the meaningful structure of source code.	Abstract, syntactic hierarchy	Calling it token list
Why is it called abstract?	It omits unnecessary grammar and punctuation details.	Removes concrete syntax noise	Saying it is vague or incomplete
AST vs parse tree?	Parse tree follows grammar fully; AST is simplified.	Concrete vs abstract	Saying they are identical

Question	Clear Answer	Expected Key Points	Common Mistakes
Draw AST for $a + b * c$.	Root is $+$; right child is $*$.	Precedence captured	Making $+$ and $*$ same level
What compiler phase creates AST?	Syntax analysis/parsing usually creates it.	After lexical analysis	Saying semantic analysis creates tokens
What uses AST?	Semantic analyzer, optimizer, code generator, IDE tools.	Traversal, annotations	Saying only parser uses it
Can AST store types?	Yes, after semantic analysis it may become annotated AST.	Type info, symbol links	Saying AST can only store syntax
How are parentheses represented?	Often not represented unless semantically needed.	Structure captures grouping	Always adding parentheses nodes
How do you traverse an AST?	Commonly with recursive traversal or visitor pattern.	Pre-order/post-order depending task	Traversing tokens instead
Is AST machine independent?	Yes, AST represents source structure, not target CPU details.	High-level IR	Calling it assembly-level

7. Deep-Dive Questions

Question	Answer
Why may AST be unsuitable for low-level optimization?	AST is high-level and tree-shaped. It does not naturally represent shared subexpressions, control-flow joins, or low-level instruction constraints.
How does AST help type checking?	Each expression node can be visited, child types can be computed, and the parent node can be validated and annotated with its resulting type.
What is an AST visitor?	A design pattern where separate visitor functions handle different node types. It keeps traversal logic organized for tasks like type checking or code generation.
Can two different source programs have the same AST?	Yes, if differences are only concrete syntax, such as redundant parentheses or whitespace.
Why do refactoring tools need ASTs?	They need structural understanding. Text search cannot distinguish variables, comments, strings, and different scopes.

8. Comparison Tables

Parse Tree vs AST

Feature	Parse Tree	AST
Also called	Concrete syntax tree	Abstract syntax tree

Feature	Parse Tree	AST
Contains grammar nonterminals	Yes	Usually no
Contains punctuation	Often yes	Usually no
Size	Larger	Smaller
Best for	Proving grammar derivation	Semantic analysis and translation
Example nodes	Expr, Term, Factor	Assign, BinaryOp, Return

Tokens vs AST

Feature	Tokens	AST
Structure	Flat list	Hierarchical tree
Produced by	Lexer	Parser
Knows precedence	No	Yes
Good for	Lexical analysis	Program understanding

9. Common Mistakes

- Confusing AST with parse tree.
- Drawing expression trees without respecting precedence.
- Thinking AST must include every semicolon and parenthesis.
- Forgetting AST can be annotated with semantic information.
- Saying AST directly represents control-flow edges like a CFG.
- Assuming AST is the final IR used for all optimizations.

10. Edge Cases / Special Cases

- Parentheses may affect AST structure but may not appear as nodes.
- Some languages preserve comments and formatting in a concrete syntax tree for tooling.
- Macros can change what AST gets generated.
- Error-tolerant parsers may build partial ASTs for IDEs.
- Desugared ASTs may replace high-level syntax with simpler forms.

11. How to Explain in Interview

"An AST is a simplified tree representation of source code. It keeps meaningful constructs like operators, assignments, loops, and function declarations, while removing grammar details such as punctuation. Compilers use ASTs for semantic analysis and as a starting point for IR generation."

12. Quick Revision Notes

- AST = meaningful syntax tree.
- Parser usually builds it.
- Smaller than parse tree.

- Captures precedence and nesting.
- Used by compilers, IDEs, linters, and transpilers.
- Interview trap: parse tree and AST are not the same.

13. Practice Tasks

1. Draw AST for $x = (a + b) * (c - d)$.
2. Compare parse tree and AST for $id + id * id$.
3. Write a recursive traversal that prints expression nodes in postorder.
4. Identify which AST nodes are needed for **if**, **while**, **return**, and function calls.
5. Explain why text replacement is unsafe for rename refactoring.

14. Final Cheat Sheet

Item	Notes
Core definition	Tree of meaningful source-code structure
Why it matters	Enables semantic analysis, tooling, and IR generation
Most asked questions	AST vs parse tree, draw AST, where used
Common comparisons	Tokens vs AST, parse tree vs AST
One-line answer	AST is the compiler-friendly tree form of source code after parsing.

3. Three-Address Code

1. Overview

Definition: Three-address code (TAC) is an intermediate representation where each instruction has at most three addresses: usually two operands and one result.

Typical form:

```
x = y op z
```

Here **x**, **y**, and **z** are addresses. They may be variable names, constants, or temporaries.

Why it matters:

- It breaks complex expressions into simple steps.
- It is easy to optimize.
- It maps naturally to many machine instructions.
- It is a common exam and interview topic because expression translation becomes mechanical.

Where it is used in real systems:

- Educational compilers use TAC heavily.

- Production compilers use TAC-like IRs.
- Optimizers use similar simple operations internally.
- Virtual machines and JIT compilers often operate on linear IR close to TAC.

Why interviewers ask about it: TAC is easy to test in interviews. They can ask you to convert expressions, loops, conditionals, array access, and function calls into intermediate code.

2. Core Idea

TAC converts one complex operation into a sequence of simple operations.

Intuition: A CPU usually cannot execute `x = a + b * c - d / e` as one instruction. TAC makes every intermediate result explicit.

Real-world analogy: Solving a math expression on paper. You do multiplication first, store the result, do division, store the result, then combine them.

Small example:

```
x = a + b * c;
```

TAC:

```
t1 = b * c
t2 = a + t1
x  = t2
```

Step-by-step:

1. Identify highest-precedence operation: `b * c`.
2. Store it in temporary `t1`.
3. Add `a + t1`.
4. Store in temporary `t2`.
5. Assign `t2` to `x`.

3. Important Subtopics

Subtopic	What It Means	Why It Matters	Example	Common Interview Angle
Temporaries	Compiler-created variables	Store intermediate results	<code>t1 = b * c</code>	"How many temporaries are needed?"
Assignment	Copy one value to another	Basic operation	<code>x = t1</code>	"Translate assignment statement."

Subtopic	What It Means	Why It Matters	Example	Common Interview Angle
Binary operations	Two operands and result	Core of TAC	<code>t1 = a + b</code>	"Convert expression to TAC."
Unary operations	One operand and result	Needed for negation, not	<code>t1 = -a</code>	"Handle unary minus."
Conditional jumps	Branch based on condition	Represents <code>if</code> and loops	<code>if x < y goto L1</code>	"Translate if-else."
Unconditional jumps	Direct control transfer	Represents loop back edges	<code>goto L2</code>	"Translate while."
Labels	Named instruction positions	Jump targets	<code>L1:</code>	"Where do labels go?"
Function calls	Parameter passing and call result	Represents procedures	<code>param x, t1 = call f, 1</code>	"Translate function call."
Array access	Address calculation	Important for memory layout	<code>t1 = i * width</code>	"Translate <code>a[i]</code> ."

4. Real-World Example

Backend server code compiled by a JIT: Suppose a Java method computes a price:

```
total = base + tax * quantity;
```

A JIT compiler may lower it into simple IR operations similar to TAC, optimize common subexpressions, remove redundant loads, and emit machine instructions.

5. Diagrams / Mental Models

Expression tree

```
      +
     / \
    a  *
      / \
     b  c
```

TAC sequence

```
t1 = b * c
t2 = a + t1
x  = t2
```

Common TAC Forms

Form	Meaning
<code>x = y op z</code>	Binary operation

Form	Meaning
<code>x = op y</code>	Unary operation
<code>x = y</code>	Assignment
<code>goto L</code>	Unconditional jump
<code>if x relop y goto L</code>	Conditional branch
<code>param x</code>	Pass argument
<code>x = call f, n</code>	Function call returning value
<code>return x</code>	Return from function

6. Common Interview Questions

Question	Clear Answer	Expected Key Points	Common Mistakes
What is TAC?	IR where each instruction has at most three addresses.	Result, operand1, operand2	Saying exactly three always
Convert <code>x=a+b*c</code> .	<code>t1=b*c; t2=a+t1; x=t2</code> .	Precedence, temporaries	Doing addition first
Why use temporaries?	To hold intermediate results explicitly.	Simplifies optimization/codegen	Reusing source variables incorrectly
What are addresses in TAC?	Names, constants, temporaries, memory locations.	Not only memory addresses	Thinking address means pointer only
How are conditionals represented?	With conditional jumps and labels.	<code>if cond goto, goto</code>	Keeping high-level <code>if</code> unchanged
How are loops represented?	Labels, condition checks, body, back jump.	Loop header and exit label	Missing back edge
Is TAC machine code?	No, it is intermediate and usually machine independent.	IR not final binary	Calling it assembly
How are function calls represented?	Using <code>param</code> , <code>call</code> , and return value temporaries.	Calling convention abstracted	Ignoring argument count/order
How is array indexing handled?	Compute offset then load/store.	Element width, base address	Forgetting element size
What are common TAC representations?	Quadruples, triples, indirect triples.	Storage formats	Confusing TAC with quadruple only

7. Deep-Dive Questions

Question	Answer
Why does TAC make optimization easier?	Each instruction has a small, explicit effect. Optimizations like constant folding, common subexpression elimination, copy propagation, and dead-code elimination become easier to implement.
How would you generate TAC from an AST?	Do a postorder traversal. Generate code for children first, store their results in temporaries, then emit the parent operation.
How does TAC represent short-circuit boolean expressions?	It usually uses conditional jumps so evaluation can skip the right side when the answer is already known.
What is the difference between TAC and SSA?	TAC allows variables to be assigned multiple times. SSA is usually TAC-like but each variable version is assigned exactly once.
Can TAC be optimized before machine code generation?	Yes. Many machine-independent optimizations are performed on TAC or TAC-like IR.

8. Comparison Tables

TAC vs AST

Feature	AST	TAC
Shape	Tree	Linear instruction sequence
Level	Higher-level	Lower-level
Temporaries	Usually implicit	Explicit
Good for	Semantic analysis	Optimization and code generation
Control flow	Nested constructs	Labels and jumps

TAC vs Assembly

Feature	TAC	Assembly
Machine dependent	Usually no	Yes
Registers	Abstract temporaries	Real or virtual registers
Instruction format	Simple 3-address style	Target-specific
Used by	Compiler middle end	Assembler/CPU path

9. Common Mistakes

- Thinking TAC instructions must always contain exactly three addresses.
- Ignoring operator precedence during conversion.
- Forgetting labels for control-flow constructs.
- Treating temporaries as real source variables.

- Forgetting array element width in address calculation.
- Confusing TAC form with its storage representation.

10. Edge Cases / Special Cases

- Unary operations use fewer than three addresses.
- Function calls may require multiple `param` instructions.
- Boolean expressions may be translated using numeric values or control flow.
- Array access differs for row-major and column-major layout.
- Object field access may require offset lookup in class layout.

11. How to Explain in Interview

"Three-address code is an intermediate code form where each instruction performs one simple operation using at most two operands and one result. It converts complex expressions into temporary-based steps, which makes optimization and target-code generation easier."

12. Quick Revision Notes

- TAC form: `x = y op z`.
- Uses temporaries like `t1`, `t2`.
- Control flow uses labels and jumps.
- Function calls use `param` and `call`.
- Interview trap: "three-address" means at most three addresses, not exactly three.

13. Practice Tasks

1. Generate TAC for `x = (a + b) * (c - d)`.
2. Generate TAC for `if (a < b) x = 1; else x = 2;`.
3. Generate TAC for `while (i < n) i = i + 1;`.
4. Generate TAC for `a[i] = b[j] + 1`.
5. Apply constant folding to TAC for `x = 2 * 3 + y`.

14. Final Cheat Sheet

Item	Notes
Core definition	IR with at most two operands and one result per instruction
Why it matters	Simple, optimizable, close to machine operations
Most asked questions	Convert expression, translate if/while, array access
Common comparisons	TAC vs AST, TAC vs assembly
One-line answer	TAC breaks complex source constructs into simple temporary-based instructions.

4. Quadruples

1. Overview

Definition: A **quadruple** is a way to represent three-address code using four fields:

```
(operator, argument1, argument2, result)
```

For example:

```
t1 = b * c
```

can be stored as:

```
(*, b, c, t1)
```

Why it matters:

- It gives TAC a structured table format.
- It is easy for compiler algorithms to process.
- It makes results explicit.
- It supports optimization passes that rewrite operands and results.

Where it is used in real systems:

- Educational compilers use quadruple tables.
- Optimizer passes often store IR instructions in records similar to quadruples.
- Static analyzers store operations in structured forms.

Why interviewers ask about it: It is a standard representation question after TAC. Interviewers often ask you to produce quadruples for an expression and compare them with triples.

2. Core Idea

Quadruples store every TAC instruction as a row in a table.

Intuition: TAC is the sentence; quadruple is the spreadsheet row.

Real-world analogy: A bank transaction can be written as text, but databases store it in columns like operation, source account, destination account, amount. Quadruples do that for compiler instructions.

Small example:

```
x = a + b * c;
```

TAC:

```
t1 = b * c
t2 = a + t1
x  = t2
```

Quadruples:

No.	op	arg1	arg2	result
0	*	b	c	t1
1	+	a	t1	t2
2	=	t2	-	x

Step-by-step:

- 1. Create a row for each TAC instruction.
- 2. Put the operation in **op**.
- 3. Put operands in **arg1** and **arg2**.
- 4. Put the destination in **result**.

3. Important Subtopics

Subtopic	What It Means	Why It Matters	Example	Common Interview Angle
Operator field	Operation being performed	Tells compiler action	+, *, goto	"What goes in op?"
Argument fields	Input operands	Data dependencies	a, b, t1	"How represent unary op?"
Result field	Destination	Makes definitions explicit	t2	"Why quadruples are easy to optimize?"
Temporaries	Named intermediate results	Referenced by later rows	t1	"Quadruples vs triples."
Jump quadruples	Branch representation	Handles control flow	(if<, a, b, L1)	"Represent if statement."
Call quadruples	Function-call rows	Handles procedures	(param, x, -, -)	"Represent function call."

4. Real-World Example

Compiler optimization pass: Suppose an optimizer sees:

op	arg1	arg2	result
+	2	3	t1

It can replace this row with:

op	arg1	arg2	result
=	5	-	t1

This is constant folding. Quadruples make the rewrite straightforward because fields are explicit.

5. Diagrams / Mental Models

TAC instruction: `t1 = a + b`

Quadruple row:

+	+	+	+	+
op	arg1	arg2	result	
+	+	+	+	+
+	a	b	t1	
+	+	+	+	+

Control-Flow Quadruple Examples

Source	Quadruple
<code>goto L1</code>	<code>(goto, -, -, L1)</code>
<code>if a < b goto L1</code>	<code>(if<, a, b, L1)</code>
<code>x = y</code>	<code>(=, y, -, x)</code>
<code>return x</code>	<code>(return, x, -, -)</code>

6. Common Interview Questions

Question	Clear Answer	Expected Key Points	Common Mistakes
What is a quadruple?	A 4-field representation of TAC: op, arg1, arg2, result.	Structured TAC	Saying it has four operands
Convert <code>x=a+b*c</code> to quadruples.	Rows for <code>*</code> , <code>+</code> , and assignment.	Temporaries explicit	Skipping assignment row
Why are quadruples useful?	They make operands and result explicit for optimization.	Easy rewriting	Saying only memory saving
What is the result field?	Destination of the operation.	Defines variable/temp	Confusing with arg2
How represent unary minus?	<code>(-, a, -, t1)</code> or <code>(uminus, a, -, t1)</code> .	arg2 empty	Forcing two operands

Question	Clear Answer	Expected Key Points	Common Mistakes
How represent assignment?	(=, source, -, destination).	Copy operation	Putting destination in arg1
Quadruples vs triples?	Quadruples name temporary results; triples refer to instruction positions.	Explicit result vs positional result	Saying both are identical
Are labels stored in quadruples?	Yes, jump targets can be labels or instruction numbers.	Control flow	Ignoring branches
Do quadruples require temporaries?	Usually yes for intermediate values.	t1, t2	Thinking no temporaries exist
Can quadruples represent calls?	Yes, using rows like param, call, and assignment.	Procedure representation	Treating calls as source syntax only

7. Deep-Dive Questions

Question	Answer
Why are quadruples easier to rearrange than triples?	Because later instructions refer to named temporaries, not instruction positions. Reordering does not automatically break references.
What is the memory cost of quadruples?	They store an explicit result field, so they may use more space than triples. The advantage is simpler optimization and code motion.
How can conditional jumps be stored as quadruples?	The relational operator and operands are stored with the target label as result, for example (if<, a, b, L1).
How do quadruples help common subexpression elimination?	The compiler can compare op, arg1, and arg2 fields to detect repeated computations and reuse an existing result.
Can quadruples represent SSA?	Yes, an SSA-style IR can use quadruple-like instruction records where each result name is assigned only once.

8. Comparison Tables

Quadruples vs TAC Text

Feature	TAC Text	Quadruple
Format	Human-readable instruction	Table/record
Example	t1 = a + b	(+, a, b, t1)
Easier for humans	Yes	Medium
Easier for algorithms	Medium	Yes

Quadruples vs Triples

Feature	Quadruples	Triples
Fields	op, arg1, arg2, result	op, arg1, arg2
Temporary names	Explicit	Often omitted
References	By temp name	By instruction number
Code movement	Easier	Harder unless indirect triples used
Space	More	Less

9. Common Mistakes

- Saying quadruples have four operands.
- Mixing up `arg2` and `result`.
- Forgetting that unary operations leave one argument field empty.
- Not using temporaries for intermediate results.
- Confusing quadruple representation with the TAC concept itself.

10. Edge Cases / Special Cases

- For `x = y`, `arg2` is unused.
- For `goto L`, both argument fields may be unused.
- Function calls may require several rows, not one row.
- Some compilers store instruction numbers instead of symbolic labels.
- Operators may be normalized, such as `if<` instead of separate `if` and `<`.

11. How to Explain in Interview

"A quadruple is a table representation of three-address code with four fields: operator, first argument, second argument, and result. It is easy for compiler passes to process because every instruction's inputs and output are explicit."

12. Quick Revision Notes

- Format: `(op, arg1, arg2, result)`.
- Best feature: explicit result.
- Easier than triples for code movement.
- Uses temporaries for intermediate values.
- Interview trap: quadruple means four fields, not four operands.

13. Practice Tasks

1. Write quadruples for `x = (a + b) * (c - d)`.
2. Write quadruples for `if a < b then x = 1 else x = 2`.
3. Represent unary minus `x = -a + b`.
4. Compare quadruple and triple representation for the same expression.
5. Identify constant-folding opportunities in a quadruple table.

14. Final Cheat Sheet

Item	Notes
Core definition	Four-field storage format for TAC
Why it matters	Explicit operands and result simplify optimization
Most asked questions	Convert expression, compare with triples
Common comparisons	Quadruples vs triples, quadruples vs TAC text
One-line answer	Quadruples store each TAC instruction as <code>(op, arg1, arg2, result)</code> .

5. Triples

1. Overview

Definition: A **triple** is a representation of three-address code using three fields:

```
(operator, argument1, argument2)
```

Unlike quadruples, triples do not store an explicit result field. The result of an instruction is referred to by the instruction's position number.

Why it matters:

- It reduces the need for temporary variable names.
- It shows how intermediate results can be referenced indirectly.
- It is a common comparison topic with quadruples.

Where it is used in real systems:

- Teaching compilers and compiler textbooks use triples to explain IR storage choices.
- Some internal IRs use instruction references instead of explicit temporary names.
- Dataflow systems often refer to operation results by node or instruction identity.

Why interviewers ask about it: Triples test whether you understand that TAC is the idea, while quadruples and triples are storage representations.

2. Core Idea

In triples, the result of each instruction is implicit. If instruction 0 computes `b * c`, later instructions refer to `(0)`.

Intuition: Instead of naming every intermediate result `t1`, `t2`, use the row number as the name.

Real-world analogy: In a spreadsheet, a formula may refer to cell `A1` instead of giving that value a separate name. Triples refer to instruction positions similarly.

Small example:

```
x = a + b * c;
```

Triples:

No.	op	arg1	arg2
0	*	b	c
1	+	a	(0)
2	=	(1)	x

Instruction 1 uses the result of instruction 0.

Step-by-step:

- 1. Compute `b * c` at row 0.
- 2. Use `(0)` as an operand in row 1.
- 3. Assign result `(1)` to `x`.

3. Important Subtopics

Subtopic	What It Means	Why It Matters	Example	Common Interview Angle
Instruction position	Row number identifies result	Removes temp names	(0)	"How are results referenced?"
Positional reference	Operand can refer to previous row	Encodes dependencies	arg2 = (0)	"What does (i) mean?"
Code motion problem	Reordering rows changes positions	Makes optimization harder	Moving row 0	"Why triples are harder to optimize?"
Indirect triples	Separate pointer table to triples	Allows reordering without changing references	pointer list	"How to fix code motion issue?"
Assignment in triples	Store source and destination in fields	No explicit result column	(=, (1), x)	"How represent assignment?"

4. Real-World Example

Expression optimizer: A compiler can represent expression computations as triples and refer to previous computations by instruction number. But if it wants to move an instruction out of a loop, positional references can become inconvenient. This motivates indirect triples or named temporaries.

5. Diagrams / Mental Models

Quadruple style:

```
t1 = b * c
t2 = a + t1
x  = t2
```

Triple style:

```
(0) * b c
(1) + a (0)
(2) = (1) x
```

Indirect Triple Mental Model

Pointer table:	Triple table:
P0 -> T0	T0: (*, b, c)
P1 -> T1	T1: (+, a, (0))
P2 -> T2	T2: (=, (1), x)

To reorder execution, change the pointer table instead of moving triple rows.

6. Common Interview Questions

Question	Clear Answer	Expected Key Points	Common Mistakes
What is a triple?	A TAC representation with op, arg1, arg2; result is instruction position.	Three fields, implicit result	Saying it has no result at all
How is result referenced?	By instruction number, like (0).	Positional reference	Inventing temporaries unnecessarily
Convert $x=a+b*c$ to triples.	(*, b, c), (+, a, (0)), (=, (1), x).	Correct dependencies	Wrong row references
Triples vs quadruples?	Quadruples have explicit result; triples use instruction position.	Space vs code motion	Saying triples are always better
Why are triples compact?	They avoid storing temporary result names.	Fewer fields	Ignoring reference overhead
What is a disadvantage of triples?	Code reordering is harder because references depend on positions.	Optimization issue	Saying no disadvantage
What are indirect triples?	Triples accessed through a pointer list to allow reordering.	Solves movement issue	Confusing with pointers in C
Can triples represent unary ops?	Yes, second argument may be empty.	Same as TAC	Forcing arg2

Question	Clear Answer	Expected Key Points	Common Mistakes
Can triples represent control flow?	Yes, jumps can be stored as triple operations.	Labels or instruction refs	Thinking only expressions
Are triples machine code?	No, they are an IR storage representation.	Compiler middle end	Calling them CPU instructions

7. Deep-Dive Questions

Question	Answer
Why does code motion become difficult with triples?	If an instruction is moved, its position number changes. Any operand referring to that position may need updating.
How do indirect triples solve code motion?	They keep triples in a stable table and reorder a separate list of pointers. References can remain stable while execution order changes.
Are triples better than quadruples?	Not always. Triples save space but make optimization and reordering less convenient. Quadruples are often simpler to manipulate.
How are dependencies represented in triples?	A dependency is represented when an argument refers to a previous instruction number.
Why might modern compilers prefer named SSA values over textbook triples?	Named SSA values make data dependencies explicit and make transformations easier while still avoiding multiple assignments to the same variable.

8. Comparison Tables

Triples vs Indirect Triples

Feature	Triples	Indirect Triples
Result reference	Instruction position	Stable triple reference through pointer table
Code movement	Harder	Easier
Extra table	No	Yes
Space	Lower	Slightly higher
Best for	Simple representation	Optimizing compilers

Triples vs Quadruples

Feature	Triples	Quadruples
Fields	3	4

Feature	Triples	Quadruples
Result	Implicit	Explicit
Temporaries	Reduced	Common
Reordering	Harder	Easier
Readability	Lower	Higher

9. Common Mistakes

- Thinking triples cannot represent intermediate results.
- Forgetting that the row number acts as the result.
- Reordering triples without updating references.
- Confusing (0) with a constant zero.
- Saying triples are the same as syntax trees.

10. Edge Cases / Special Cases

- Forward references are usually avoided in expression triples.
- Labels may still be symbolic rather than numeric.
- Indirect triples introduce a pointer table.
- Some operations need only one argument.
- Assignment triples may look unusual because destination is stored as an argument.

11. How to Explain in Interview

"A triple represents a TAC instruction using only operator and operands. The result is not stored in a separate field; it is identified by the instruction number. This saves temporary names but makes code movement harder."

12. Quick Revision Notes

- Format: $(op, arg1, arg2)$.
- Result is instruction index.
- (0) means result of instruction 0.
- Main disadvantage: hard to reorder.
- Indirect triples use pointer tables.

13. Practice Tasks

1. Generate triples for $x = (a + b) * (c - d)$.
2. Convert a quadruple table into triples.
3. Show how moving one triple can break references.
4. Draw an indirect triple table for a three-instruction expression.
5. Compare space and optimization convenience of triples and quadruples.

14. Final Cheat Sheet

Item	Notes
Core definition	Three-field TAC representation with implicit result
Why it matters	Saves temp names and teaches IR storage trade-offs
Most asked questions	Triples vs quadruples, result references
Common comparisons	Triples vs quadruples, triples vs indirect triples
One-line answer	A triple stores (op, arg1, arg2) and uses the instruction number as the result.

6. Control-Flow Graph

1. Overview

Definition: A **control-flow graph (CFG)** is a directed graph that represents all possible execution paths through a program or function.

In a CFG:

- Nodes are usually **basic blocks**.
- Edges represent possible flow of control from one block to another.

Why it matters:

- It is essential for analyzing branches, loops, reachability, and optimization.
- Many compiler optimizations are performed over CFGs.
- It is used in static analysis, security analysis, testing, and performance tuning.

Where it is used in real systems:

- Compilers use CFGs for dataflow analysis and optimization.
- Static analyzers use CFGs to detect unreachable code and possible null dereferences.
- Test coverage tools reason about branches and paths.
- Security tools use CFGs to analyze suspicious control transfers.

Why interviewers ask about it: CFGs connect compiler design with graph algorithms. They test whether you can reason about execution paths instead of just syntax.

2. Core Idea

A CFG shows where execution can go next.

Intuition: An AST shows nesting. A CFG shows movement.

Real-world analogy: A road map. Intersections are basic blocks, roads are possible transitions. A loop is a road that leads back to an earlier intersection.

Small example:

```
if (x > 0)
    y = 1;
else
    y = 2;
z = y + 3;
```

CFG:



Step-by-step:

- 1. Create a block for the condition.
- 2. Add one edge for true and one for false.
- 3. Add blocks for then and else bodies.
- 4. Join both branches at the following statement.

3. Important Subtopics

Subtopic	What It Means	Why It Matters	Example	Common Interview Angle
Nodes	Basic blocks	Units of sequential execution	B1, B2	"What does a CFG node represent?"
Edges	Possible control transfers	Represent branches and loops	true/false edge	"What do CFG edges mean?"
Entry block	Function starting point	Analysis starts here	first block	"What is entry node?"
Exit block	Function return/end	Analysis may collect results here	return block	"Multiple returns?"
Dominators	A node dominates another if every path to it passes through the dominator	Used in SSA and optimization	entry dominates all	"What is a dominator?"

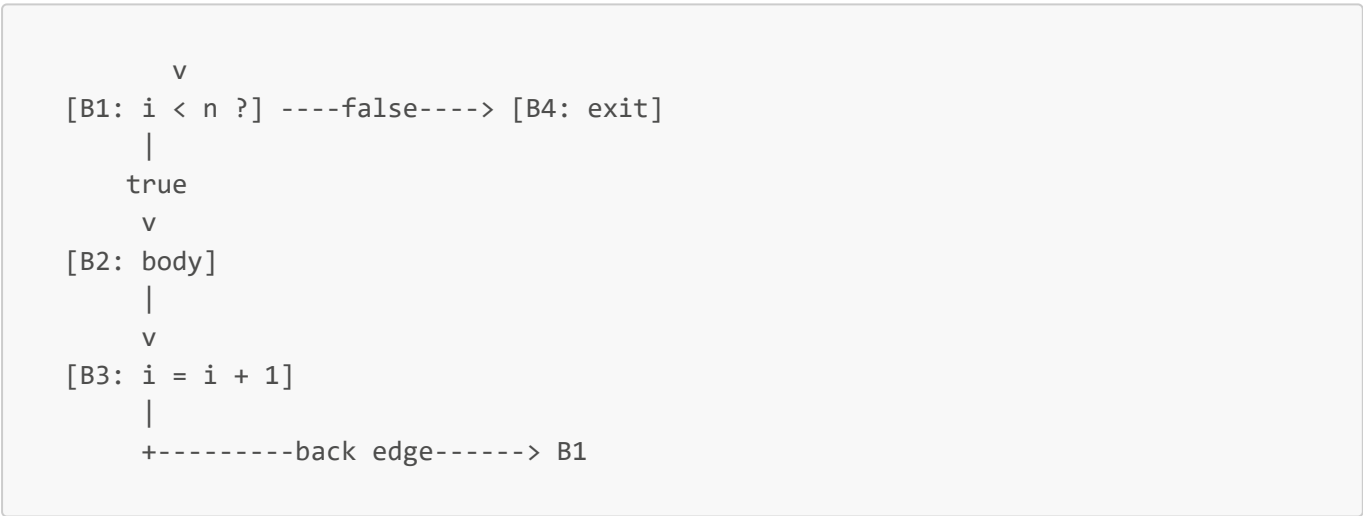
Subtopic	What It Means	Why It Matters	Example	Common Interview Angle
Back edges	Edges that point to earlier/dominating block	Identify loops	loop body to header	"How detect loops?"
Reachability	Whether a block can execute	Remove dead code	block after <code>return</code>	"Find unreachable code."
Dataflow analysis	Facts propagated over CFG	Powers optimizations	live variables	"Why CFG needed?"

4. Real-World Example

Static analyzer detecting missing return: In a function with several `if` branches, a CFG can show whether every path reaches a `return`. If one path reaches the end without returning a value, the compiler or analyzer can report an error.

5. Diagrams / Mental Models

While Loop CFG



CFG vs AST Mental Model

Program View	Best Question Answered
AST	"What is nested inside what?"
CFG	"What can execute after this?"
Dataflow graph	"Which value depends on which value?"

6. Common Interview Questions

Question	Clear Answer	Expected Key Points	Common Mistakes
What is a CFG?	A directed graph of possible execution paths.	Nodes as basic blocks, edges as control flow	Saying it is expression tree
What are CFG nodes?	Usually basic blocks.	Straight-line code blocks	Saying every token
What are CFG edges?	Possible transfers of control.	Branches, fall-through, loop back edges	Forgetting false edge
Why is CFG needed?	To analyze branches, loops, reachability, and dataflow.	Optimizations depend on it	Saying only for diagrams
Draw CFG for if-else.	Condition node splits into true/false blocks and rejoins.	Join block	Missing join
Draw CFG for while loop.	Header condition, body, back edge, exit edge.	Back edge	Drawing no exit
What is unreachable code?	Code with no path from entry.	Can be removed/warned	Confusing with dead assignment
What is a back edge?	Edge from a block to a loop header/earlier dominator.	Loop detection	Any backward arrow
What is an entry block?	First block where function execution begins.	Analysis start	Multiple arbitrary starts
How does CFG help optimization?	Dataflow analyses run over CFG to find constants, liveness, dead code.	Global reasoning	Only local expression optimization

7. Deep-Dive Questions

Question	Answer
What is a dominator?	Block A dominates block B if every path from the entry block to B must pass through A. Dominators are used in loop analysis and SSA construction.
How are loops identified in CFGs?	Back edges often indicate natural loops, especially when the target dominates the source.
Why are basic blocks used as CFG nodes instead of individual instructions?	Basic blocks reduce graph size because all instructions inside a block execute sequentially without internal branching.
How does CFG support liveness analysis?	Liveness facts propagate backward along CFG edges to determine whether variable values may be used in the future.

Question	Answer
How can exceptions affect CFGs?	A statement that may throw can have exceptional edges to handler blocks, making the CFG more complex.

8. Comparison Tables

AST vs CFG

Feature	AST	CFG
Structure	Tree	Directed graph
Represents	Syntax nesting	Execution paths
Loop representation	Loop node	Back edge
Branch representation	If node	Multiple outgoing edges
Best for	Semantic analysis, syntax transforms	Dataflow, optimization

CFG vs Dataflow Graph

Feature	CFG	Dataflow Graph
Edges mean	Control can move	Value depends on another value
Focus	Execution order	Data dependencies
Nodes	Basic blocks/instructions	Operations/values
Used for	Reachability, liveness	Scheduling, expression optimization

9. Common Mistakes

- Drawing CFG as a tree and forgetting joins.
- Missing the false edge from a condition.
- Missing loop back edges.
- Confusing dead code with unreachable code.
- Thinking CFG only exists after machine code generation.
- Ignoring fall-through edges.

10. Edge Cases / Special Cases

- `break` and `continue` create edges to loop exit or loop header.
- `return` blocks usually go to function exit.
- Exceptions add hidden edges.
- `switch` can create multiple outgoing edges.
- Short-circuit boolean expressions create control-flow edges.
- Infinite loops may not have an exit edge.

11. How to Explain in Interview

"A control-flow graph is a directed graph showing all possible execution paths in a function. Its nodes are usually basic blocks and its edges represent jumps, branches, fall-throughs, and loop back edges. Compilers use CFGs for dataflow analysis and optimization."

12. Quick Revision Notes

- CFG = execution-path graph.
- Nodes = basic blocks.
- Edges = possible control transfers.
- Loops create back edges.
- Branches split; joins merge.
- Interview trap: AST shows structure, CFG shows execution flow.

13. Practice Tasks

1. Draw CFG for an if-else program.
2. Draw CFG for a while loop with **break**.
3. Identify unreachable blocks in a TAC sequence.
4. Mark back edges in a loop CFG.
5. Run liveness analysis manually on a four-block CFG.

14. Final Cheat Sheet

Item	Notes
Core definition	Directed graph of possible execution paths
Why it matters	Enables dataflow analysis, optimization, reachability
Most asked questions	Draw CFG, CFG vs AST, loops, back edges
Common comparisons	AST vs CFG, CFG vs dataflow graph
One-line answer	A CFG shows how control can move between basic blocks during execution.

7. Basic Blocks

1. Overview

Definition: A **basic block** is a maximal sequence of consecutive instructions with:

1. One entry point, at the first instruction.
2. One exit point, at the last instruction.
3. No internal jumps into or out of the middle.

If the first instruction executes, every instruction in the block executes in order.

Why it matters:

- Basic blocks are the building units of CFGs.

- Many optimizations are first applied within basic blocks.
- They simplify reasoning about control flow.
- They are essential for code generation and instruction scheduling.

Where it is used in real systems:

- Compilers group IR instructions into basic blocks.
- CPU branch prediction and instruction scheduling often reason about blocks.
- Static analyzers use blocks to reduce graph complexity.
- Profilers report hot blocks or hot paths.

Why interviewers ask about it: Basic block construction is a classic compiler-design problem. Interviewers may give TAC and ask you to identify leaders and blocks.

2. Core Idea

A basic block is a straight-line region of code.

Intuition: Once you enter a basic block, there are no choices until the end.

Real-world analogy: A hallway with one entrance and one exit. Once inside, you walk straight through; decisions happen only at intersections between hallways.

Small example:

```
1: t1 = a + b
2: t2 = t1 * c
3: if t2 < d goto 6
4: x = t2
5: goto 7
6: x = d
7: return x
```

Leaders:

- 1: first instruction
- 4: instruction after conditional jump
- 6: jump target
- 7: jump target and instruction after goto

Basic blocks:

```
B1: 1, 2, 3
B2: 4, 5
B3: 6
B4: 7
```

3. Important Subtopics

Subtopic	What It Means	Why It Matters	Example	Common Interview Angle
Leaders	First instruction of a basic block	Used to partition TAC	first instruction, jump target	"Find leaders."
Maximal sequence	Include as many instructions as possible	Avoid unnecessary blocks	straight-line instructions	"Can this be same block?"
Fall-through	Control naturally moves to next block	Adds CFG edge	after condition false	"Add edges."
Jump target	Instruction reached by branch/goto	Starts new block	label L1	"Why target is leader?"
Local optimization	Optimization within one block	Simpler than global	constant folding	"Local vs global optimization."
Basic block DAG	DAG built for one block	Finds common subexpressions	a+b reused	"DAG representation."

4. Real-World Example

Compiler removes redundant computation inside a block:

```
t1 = a + b
t2 = a + b
x  = t1 + t2
```

Inside a basic block, the compiler can see **a + b** is computed twice without changes to **a** or **b**, so it can reuse **t1**.

5. Diagrams / Mental Models

Leader Rules

Rule	Why
First instruction is a leader	Program starts there
Target of any jump is a leader	Control can enter there
Instruction after a jump is a leader	Control may continue there after branch or from elsewhere

Basic Blocks to CFG

```
Basic blocks:

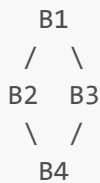
B1: condition
```

```

B2: then part
B3: else part
B4: join

```

CFG:



6. Common Interview Questions

Question	Clear Answer	Expected Key Points	Common Mistakes
What is a basic block?	A maximal straight-line sequence with one entry and one exit.	No internal jumps	Saying any group of statements
What is a leader?	First instruction of a basic block.	First instruction, jump target, after jump	Missing instruction after jump
How to form basic blocks?	Find leaders, then each leader starts a block until before next leader.	Standard algorithm	Splitting every line
Why use basic blocks?	Reduce CFG size and enable local optimization.	Sequential execution	Saying only for readability
Can a basic block contain a branch?	Yes, but only as the last instruction.	Exit at end	Branch in middle
Can control enter middle of a block?	No. If it can, that instruction must start a new block.	One entry	Ignoring jump target
What is local optimization?	Optimization within a single basic block.	No cross-block facts	Calling all optimization local
Basic block vs CFG?	Basic blocks are nodes; CFG connects them with edges.	Building blocks	Saying same thing
Why maximal?	To avoid unnecessary fragmentation.	Include all straight-line instructions	Creating tiny blocks
What ends a block?	A jump, conditional branch, return, or instruction before next leader.	Control transfer	Ending after every assignment

7. Deep-Dive Questions

Question	Answer
Why is local optimization easier than global optimization?	Inside a basic block, execution is sequential and guaranteed after entry, so the compiler does not need to reason about multiple paths.
How do basic blocks help instruction scheduling?	Instructions inside a block can often be reordered if dependencies are preserved, without changing control flow.
How are basic blocks affected by labels?	A labeled instruction is usually a leader because control can jump to it.
Can a basic block have multiple successors?	Yes. A block ending in a conditional branch usually has two successors.
Can a basic block have multiple predecessors?	Yes. A join block after an if-else may have multiple incoming edges.

8. Comparison Tables

Basic Block vs Statement

Feature	Statement	Basic Block
Unit type	Source-level construct	IR/control-flow unit
Size	Usually one source action	Multiple instructions
Entry	Source order	One control-flow entry
Exit	Source order	One final transfer

Local vs Global Optimization

Feature	Local Optimization	Global Optimization
Scope	One basic block	Across CFG
Complexity	Lower	Higher
Needs CFG?	Not always	Yes
Example	Common subexpression in one block	Loop-invariant code motion

9. Common Mistakes

- Forgetting that the first instruction is a leader.
- Missing jump targets.
- Missing the instruction after a jump.
- Ending a block too early.
- Allowing a jump into the middle of a block.
- Confusing block boundaries with source-line boundaries.

10. Edge Cases / Special Cases

- **return** ends a block.
- Unconditional **goto** ends a block; the next instruction may still be a leader.
- Conditional branch has two successors.
- A label in the middle of straight-line code starts a new block.
- Exception-throwing instructions may end or split blocks in advanced CFGs.

11. How to Explain in Interview

"A basic block is a maximal straight-line sequence of IR instructions with one entry and one exit. Once control enters the block, all instructions execute in order. CFG nodes are usually basic blocks."

12. Quick Revision Notes

- Basic block = straight-line code.
- Leaders: first instruction, jump targets, instructions after jumps.
- Branches appear only at block ends.
- Used as CFG nodes.
- Local optimization happens inside blocks.

13. Practice Tasks

1. Given TAC with labels and gotos, find leaders.
2. Divide a TAC sequence into basic blocks.
3. Draw CFG from basic blocks.
4. Find local common subexpressions in a block.
5. Identify whether a proposed block violates one-entry/one-exit rules.

14. Final Cheat Sheet

Item	Notes
Core definition	Maximal straight-line instruction sequence
Why it matters	Forms CFG nodes and enables local optimization
Most asked questions	Find leaders, form blocks, draw CFG
Common comparisons	Basic block vs statement, local vs global optimization
One-line answer	A basic block is code that executes sequentially from one entry to one exit.

8. Static Single Assignment Form Basics

1. Overview

Definition: Static Single Assignment (SSA) form is an intermediate representation where each variable is assigned exactly once.

If a source variable is assigned multiple times, SSA creates different versions:

```
x1 = 10
x2 = x1 + 1
x3 = x2 * 2
```

Why it matters:

- It makes data dependencies explicit.
- It simplifies many optimizations.
- It helps compilers reason about values instead of mutable variables.
- It is widely used in modern compilers.

Where it is used in real systems:

- LLVM IR is in SSA form for register-like values.
- GCC uses SSA in its GIMPLE optimization pipeline.
- JavaScript JITs use SSA-like forms.
- Static analyzers use SSA-style representations for precise value tracking.

Why interviewers ask about it: SSA is a modern compiler concept that connects IR, CFG, dataflow, phi functions, and optimization.

2. Core Idea

SSA turns variable updates into new variable versions.

Intuition: Instead of asking "what is the current value of x?", SSA asks "which definition of x reaches this use?"

Real-world analogy: Version control for variables. Every assignment creates a new commit/version. You never overwrite an old version.

Small example:

Original:

```
x = 1;
x = x + 2;
y = x * 3;
```

SSA:

```
x1 = 1
x2 = x1 + 2
y1 = x2 * 3
```

Step-by-step:

1. Rename each assigned variable.

2. Replace later uses with the correct latest version.
3. At control-flow joins, use phi functions to choose between versions.

3. Important Subtopics

Subtopic	What It Means	Why It Matters	Example	Common Interview Angle
Single assignment	Each SSA name assigned once	Simplifies reasoning	<code>x1, x2</code>	"What does SSA mean?"
Variable versioning	Multiple source assignments become versions	Preserves updates	<code>i1, i2, i3</code>	"Convert to SSA."
Def-use chain	Links definition to uses	Optimization-friendly	<code>x2</code> used by <code>y1</code>	"Why SSA helps optimization?"
Phi function	Merges values at CFG joins	Handles branches	<code>x3 = phi(x1, x2)</code>	"Why phi needed?"
Dominance	Definitions must dominate uses	Valid SSA construction	def before all uses	"Role of dominators?"
SSA destruction	Convert SSA back to normal form	Needed for machine code	copies inserted	"How lower phi?"

4. Real-World Example

Dead code elimination in compiler: In SSA:

```
x1 = expensive()
y1 = 5
return y1
```

If `x1` has no uses and `expensive()` has no side effects, the compiler can remove it. SSA makes unused definitions easy to find.

5. Diagrams / Mental Models

Variable Versions

```
Normal variable:
x changes over time: x -> x -> x

SSA variables:
x1 = first value
```

```
x2 = second value
x3 = third value
```

SSA at Branch Join

```

      if cond
      /      \
x1 = 1    x2 = 2
      \      /
x3 = phi(x1, x2)
y1 = x3 + 1

```

6. Common Interview Questions

Question	Clear Answer	Expected Key Points	Common Mistakes
What is SSA?	IR form where each variable version is assigned once.	Single assignment, versions	Saying source variable only once
Why use SSA?	Makes data dependencies explicit and optimizations simpler.	Def-use chains	Saying only saves memory
Convert simple assignments to SSA.	Rename each assignment and update uses.	Correct versions	Using old variable names
What happens at branches?	Phi functions merge possible incoming values.	CFG joins	Ignoring branch joins
What is a phi function?	A pseudo-operation selecting value based on predecessor block.	Control-flow dependent	Treating it as runtime function call
Is SSA source code?	No, it is compiler IR.	Internal representation	Writing SSA manually in normal code
Does SSA mean variable is immutable?	SSA names are immutable, but source variables may be mutable.	Versions simulate mutation	Saying language becomes functional
How does SSA help constant propagation?	Constants are attached to specific definitions and propagated along uses.	Def-use clarity	Missing control-flow merges
How is SSA removed?	Phi functions are lowered into copies and names mapped to registers/stack.	SSA destruction	Leaving phi in machine code

Question	Clear Answer	Expected Key Points	Common Mistakes
Which compilers use SSA?	LLVM, GCC, JIT compilers.	Modern compiler pipelines	Saying only academic compilers

7. Deep-Dive Questions

Question	Answer
Why is dominance important in SSA?	A definition should dominate its uses so every path to the use has seen the definition. Phi nodes handle cases where different definitions reach a join.
How does SSA help common subexpression elimination?	Since each value name has one definition, it is easier to know whether operands have changed between two expressions.
What is SSA destruction?	It is the process of converting SSA back to a form with normal assignments by replacing phi nodes with copies in predecessor blocks.
Does memory also become SSA?	Register-like values commonly use SSA. Memory SSA is more complex because loads/stores can alias.
Why do phi nodes appear at dominance frontiers?	Dominance frontiers are places where different control-flow paths meet, so multiple definitions may reach the same use.

8. Comparison Tables

Normal TAC vs SSA TAC

Feature	Normal TAC	SSA Form
Assignment count	Variable can be assigned many times	Each version assigned once
Variable names	x reused	x1, x2, x3
Data dependencies	Less explicit	Very explicit
Branch merge	Same variable name reused	Phi function
Optimization	Harder	Easier

SSA vs Functional Programming

Feature	SSA	Functional Programming
Purpose	Compiler IR	Programming paradigm
Mutation in source	Allowed	Usually avoided
Names	Compiler-generated versions	Programmer-level values
Phi nodes	Yes	No direct equivalent in source

9. Common Mistakes

- Thinking SSA means the original program cannot assign x twice.
- Forgetting phi functions at merge points.
- Treating phi as a normal runtime function.
- Using a variable version before it is defined.
- Assuming all memory operations are automatically simple in SSA.
- Forgetting SSA must eventually be lowered before final machine code.

10. Edge Cases / Special Cases

- Loops need phi functions for variables updated inside the loop.
- Branches that assign a variable on only one path may need a phi with an old value.
- Memory aliasing makes SSA more complex.
- Phi nodes conceptually execute at block entry.
- Critical edges may need splitting during SSA destruction.

11. How to Explain in Interview

"SSA is an IR form where every variable version is assigned exactly once. Reassignments become new names like $x1$, $x2$, and values from different branches are merged using phi functions. This makes data dependencies explicit and simplifies optimizations."

12. Quick Revision Notes

- SSA = Static Single Assignment.
- Each SSA name has one definition.
- Reassignments become new versions.
- Phi functions merge versions at CFG joins.
- Used by LLVM, GCC, JITs.
- Interview trap: phi is not an ordinary source-level function call.

13. Practice Tasks

1. Convert straight-line assignments into SSA.
2. Convert an if-else assignment into SSA using phi.
3. Convert a loop counter update into SSA.
4. Identify dead SSA definitions.
5. Explain how SSA helps constant propagation.

14. Final Cheat Sheet

Item	Notes
Core definition	IR where each variable version is assigned once
Why it matters	Simplifies dataflow and optimization
Most asked questions	Convert to SSA, explain phi, SSA vs TAC

Item	Notes
Common comparisons	Normal TAC vs SSA, SSA vs functional style
One-line answer	SSA turns variable updates into distinct versions so value flow is explicit.

9. DAG Representation of Expressions

1. Overview

Definition: A **DAG representation of expressions** is a directed acyclic graph used to represent computations in an expression or basic block, where common subexpressions can be shared.

DAG stands for **Directed Acyclic Graph**:

- Directed: edges have direction.
- Acyclic: no cycles.
- Graph: nodes can be shared, unlike a tree.

Why it matters:

- It detects common subexpressions.
- It avoids redundant computation.
- It helps local optimization inside basic blocks.
- It provides a bridge between expression trees and optimized code.

Where it is used in real systems:

- Compiler optimizers detect repeated expressions.
- Query optimizers reuse repeated computations.
- Spreadsheet engines build dependency graphs.
- Build systems use DAGs for task dependencies.

Why interviewers ask about it: DAGs test whether you understand expression optimization beyond simply generating TAC.

2. Core Idea

An expression tree duplicates repeated computations. A DAG shares them.

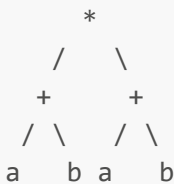
Intuition: If two parts of code compute $a + b$ and neither a nor b changed, compute it once and reuse it.

Real-world analogy: If two recipes need chopped onions, chop them once and use the result in both recipes.

Small example:

```
x = (a + b) * (a + b);
```

Expression tree duplicates $a + b$:



DAG shares $a + b$:



Optimized TAC:

```
t1 = a + b
x  = t1 * t1
```

3. Important Subtopics

Subtopic	What It Means	Why It Matters	Example	Common Interview Angle
Leaf nodes	Variables/constants	Inputs to expression	$a, 5$	"What are leaves?"
Interior nodes	Operators	Computations	$+, *$	"Draw DAG."
Common subexpression	Same operation on same unchanged operands	Can reuse result	$a + b$ repeated	"Optimize expression."
Value numbering	Assign numbers to equivalent values	Detects duplicates	same value number	"How compiler finds CSE?"
Local DAG	DAG for one basic block	Easier and safe locally	block-level optimization	"Why basic block?"
Killed variables	Assignment changes variable value	Invalidates old expressions	$a = \dots$	"When can reuse fail?"

4. Real-World Example

Database query execution: A SQL query may compute the same expression in **SELECT**, **WHERE**, and **ORDER BY**. A query optimizer can represent computations as a graph and avoid repeating expensive expressions when safe.

5. Diagrams / Mental Models

Tree vs DAG

Feature	Expression Tree	Expression DAG
Repeated expression	Duplicated	Shared
Shape	Tree	Directed acyclic graph
Optimization	Less explicit	More explicit
Node sharing	No	Yes

Basic Block DAG Example

```
t1 = a + b
t2 = a + b
t3 = t1 * t2
```

DAG:

```
      *
     / \
    v   v
   +   v
  / \
 a  b
```

6. Common Interview Questions

Question	Clear Answer	Expected Key Points	Common Mistakes
What is a DAG?	Directed acyclic graph.	Directed, no cycles	Saying it is always a tree
Why use DAG for expressions?	To share common subexpressions and avoid recomputation.	Local optimization	Only saying visualization
DAG vs expression tree?	DAG can share nodes; tree cannot.	Common subexpression	Drawing duplicates
What are leaves?	Variables or constants.	Inputs	Making operators leaves
What are interior nodes?	Operators/computations.	Computed values	Missing operands

Question	Clear Answer	Expected Key Points	Common Mistakes
What is common subexpression elimination?	Reusing a previously computed same expression.	Same operands unchanged	Reusing after variable assignment
Why is DAG acyclic?	Expression dependencies flow from operands to results without circular dependency.	No cycles in expression eval	Confusing with CFG loops
Can DAG optimize across blocks?	Basic block DAG is local; global CSE needs CFG/dataflow.	Scope matters	Assuming local DAG handles all
What kills an expression?	Assignment to one of its operands.	Value changes	Ignoring mutation
How generate TAC from DAG?	Emit code in dependency order, computing shared nodes once.	Topological order	Emitting before operands

7. Deep-Dive Questions

Question	Answer
How does a compiler detect equivalent DAG nodes?	It can use value numbering or a table keyed by operator and operand nodes. If the same key appears again and operands are unchanged, reuse the node.
Why is DAG optimization usually local in textbooks?	Within a basic block, control flow is straight-line, so it is easier to know all instructions execute in order.
What happens when <code>a</code> is reassigned?	Expressions depending on the old value of <code>a</code> cannot be reused for later computations using the new <code>a</code> .
Can algebraic identities be applied using DAGs?	Yes, but carefully. For example, <code>x + 0 = x</code> ; however floating-point and overflow rules may restrict transformations.
How is a DAG different from a CFG?	A DAG represents data dependencies in expressions; a CFG represents possible control-flow paths and may contain cycles.

8. Comparison Tables

DAG vs CFG

Feature	DAG	CFG
Represents	Data/expression dependencies	Control-flow paths
Cycles	No	Yes, loops create cycles
Nodes	Values/operators	Basic blocks
Edges	Operand dependencies	Execution transfer

Feature	DAG	CFG
Main use	Local expression optimization	Global program analysis

Common Subexpression Elimination vs Constant Folding

Feature	CSE	Constant Folding
Removes	Repeated computations	Computations with constant operands
Example	reuse $a+b$	replace $2*3$ with 6
Needs unchanged variables	Yes	Not usually
DAG helps?	Strongly	Sometimes

9. Common Mistakes

- Drawing a tree when a DAG should share nodes.
- Reusing an expression after one operand changes.
- Confusing CFG cycles with expression DAGs.
- Forgetting that local DAG optimization is limited to a basic block.
- Applying algebraic laws unsafely with floating-point arithmetic.

10. Edge Cases / Special Cases

- $a + b$ and $b + a$ may be equivalent for integer addition, but compiler must know operation properties.
- Floating-point arithmetic is not always safely associative.
- Function calls may have side effects, so repeated calls may not be reusable.
- Pointer writes can invalidate memory loads.
- Volatile variables should not be optimized away casually.

11. How to Explain in Interview

"A DAG representation shares repeated expression nodes. Unlike an expression tree, it does not duplicate common subexpressions. This lets the compiler compute expressions like $a+b$ once and reuse the value when operands have not changed."

12. Quick Revision Notes

- DAG = directed acyclic graph.
- Used for expression/basic-block optimization.
- Shares common subexpressions.
- Leaves = variables/constants.
- Interior nodes = operators.
- Interview trap: do not reuse expressions after operand mutation.

13. Practice Tasks

1. Draw DAG for $(a+b)*(a+b)$.
2. Draw DAG for $a*b + a*b + c$.

3. Convert a DAG into optimized TAC.
4. Identify killed expressions after assignment.
5. Compare DAG and CFG for a loop program.

14. Final Cheat Sheet

Item	Notes
Core definition	Directed acyclic graph representing expression dependencies
Why it matters	Detects and eliminates redundant computation
Most asked questions	DAG vs tree, CSE, generate TAC
Common comparisons	DAG vs CFG, CSE vs constant folding
One-line answer	A DAG shares equivalent expression nodes so the compiler can compute them once.

10. Phi Functions

1. Overview

Definition: A **phi function** is a special SSA operation used at a control-flow join to select the correct variable version depending on which predecessor block executed.

Example:

```
x3 = phi(x1, x2)
```

This means: if control came from one predecessor, use **x1**; if it came from another, use **x2**.

Why it matters:

- It makes SSA work with branches and loops.
- It represents merged values without losing single-assignment property.
- It is essential for SSA-based optimizations.

Where it is used in real systems:

- LLVM IR uses **phi** instructions.
- GCC SSA uses phi nodes.
- JIT compilers use phi-like merge nodes.
- Static analyzers use similar merge concepts for dataflow facts.

Why interviewers ask about it: Phi functions are the part of SSA students often memorize but do not understand. Interviewers use them to test real control-flow reasoning.

2. Core Idea

Phi functions merge different incoming versions of the same source variable.

Intuition: At a join point, the compiler needs one name for "the value of x after the if-else."

Real-world analogy: Two roads merge into one road. A sign at the merge says, "Cars from road A came from city 1; cars from road B came from city 2." Phi records which value came from which road.

Small example:

```
if (cond)
    x = 1;
else
    x = 2;
y = x + 3;
```

SSA:

```
if cond goto B1 else B2

B1:
    x1 = 1
    goto B3

B2:
    x2 = 2
    goto B3

B3:
    x3 = phi(x1 from B1, x2 from B2)
    y1 = x3 + 3
```

Step-by-step:

- 1. Each branch assigns a different SSA version.
- 2. Control paths join at B3.
- 3. Phi creates a new version representing the value after the join.
- 4. Later uses read the phi result.

3. Important Subtopics

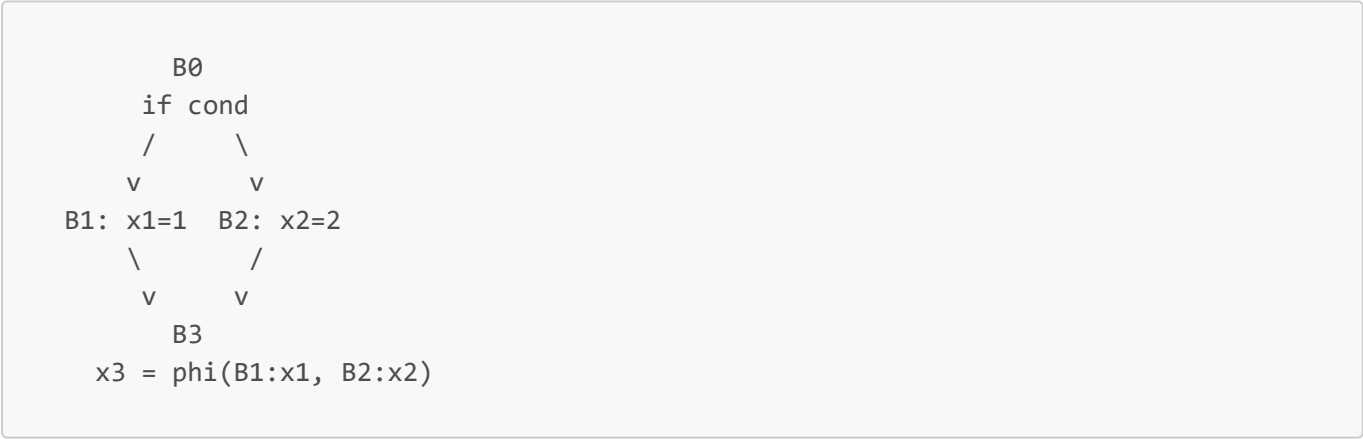
Subtopic	What It Means	Why It Matters	Example	Common Interview Angle
Join point	Block with multiple predecessors	Place where values merge	after if-else	"Where put phi?"

Subtopic	What It Means	Why It Matters	Example	Common Interview Angle
Incoming value	Value from a predecessor block	Phi depends on control path	x1 from B1	"How phi chooses?"
Phi result	New SSA version	Maintains single assignment	x3	"Why new name?"
Loop phi	Merges initial and updated loop value	Represents loop-carried dependency	i2 = phi(i0, i1)	"SSA for loops."
Phi lowering	Replace phi with copies	Required before machine code	copies in predecessors	"Does CPU execute phi?"
Parallel copy	Phi assignments happen conceptually together	Avoids ordering bugs	swap values	"Why phi lowering is tricky?"

4. Real-World Example

JIT compiler optimizing a branch: If one branch sets `type = int` and another sets `type = float`, the JIT needs a merged representation after the branch. Phi-like nodes help the optimizer reason about the possible value and type after control flow joins.

5. Diagrams / Mental Models



Loop Phi

```
i0 = 0
goto Header

Header:
  i1 = phi(i0 from Entry, i2 from Body)
  if i1 < n goto Body else Exit

Body:
```

```
i2 = i1 + 1
goto Header
```

6. Common Interview Questions

Question	Clear Answer	Expected Key Points	Common Mistakes
What is a phi function?	SSA merge operation at CFG joins.	Selects by predecessor	Calling it normal function
Why is phi needed?	To merge multiple reaching definitions while preserving single assignment.	Branch joins	Reusing same x
Where are phi nodes placed?	At blocks where different definitions can reach from different predecessors.	Join points/dominance frontier	Putting after every assignment
Does phi execute at runtime?	Not as a normal function; it is a compiler IR construct.	Lowered later	Saying CPU calls phi
What does x3=phi(x1,x2) mean?	x3 is x1 or x2 depending on incoming control edge.	Edge-specific values	Saying it picks randomly
Why phi in loops?	To merge initial value and updated value from previous iteration.	Loop-carried value	Forgetting initial value
How lower phi?	Insert copies in predecessor blocks, then remove phi.	SSA destruction	Leaving phi in assembly
Can phi have more than two inputs?	Yes, one per predecessor edge.	Multi-way joins	Assuming only if-else
Is phi same as ternary operator?	No. Ternary is source expression; phi is IR merge based on predecessor block.	Control-flow merge	Treating as cond ? a : b always
What if variable assigned only one branch?	Phi may merge new value with old incoming value.	Partial assignment	Ignoring old value

7. Deep-Dive Questions

Question	Answer
Why are phi operands associated with predecessor edges?	The selected value depends on the block from which control arrived, not on evaluating a runtime condition inside the phi itself.

Question	Answer
Why are phi nodes placed at block beginning?	They define values available to all instructions in the joined block and conceptually happen before normal instructions in that block.
What problem happens when lowering multiple phi nodes?	Phi nodes act like parallel copies. Naively ordering copies can overwrite values too early, especially in swaps.
How do phi nodes help sparse dataflow analysis?	SSA def-use chains allow analyses to follow value uses directly instead of scanning all assignments to a variable.
Are phi functions needed without branches?	Usually no. Straight-line code can be converted to SSA by simple renaming. Phi is needed when control-flow paths merge.

8. Comparison Tables

Phi Function vs Normal Assignment

Feature	Phi Function	Normal Assignment
Appears in	SSA IR	Source/IR
Meaning	Merge incoming values	Compute/copy value
Depends on	Predecessor edge	Operand evaluation
Runtime function?	No	Yes, assignment action exists
Lowered before machine code	Yes	Usually directly mapped

Phi vs Ternary Operator

Feature	Phi	Ternary
Level	Compiler IR	Source expression
Selection basis	Incoming CFG edge	Boolean condition
Location	Basic block entry	Expression position
Used for	SSA merging	Programmer expression

9. Common Mistakes

- Saying phi is a real function call.
- Forgetting one phi input per predecessor.
- Placing phi in non-join blocks unnecessarily.
- Ignoring loop phi nodes.
- Thinking phi evaluates all alternatives like normal expressions.
- Lowering phi with copies in the wrong block.

10. Edge Cases / Special Cases

- A block with three predecessors may need a phi with three inputs.
- Loops require phi for induction variables.
- If a variable is not assigned on one branch, phi must use the old version for that branch.
- Critical edges may need splitting before phi lowering.
- Multiple phi nodes in one block conceptually execute in parallel.

11. How to Explain in Interview

"A phi function is an SSA merge operation placed at a CFG join. It creates a new variable version by selecting the value that came from the actually executed predecessor block. It keeps SSA valid when branches or loops merge different definitions."

12. Quick Revision Notes

- Phi appears at CFG joins.
- One input per predecessor.
- Used in SSA.
- Not a runtime function call.
- Loops need phi for variables updated each iteration.
- Interview trap: phi selection is based on control-flow edge.

13. Practice Tasks

1. Add phi nodes to an if-else SSA example.
2. Add phi nodes to a while-loop counter.
3. Lower a simple phi into predecessor copies.
4. Explain why phi is not the same as ternary.
5. Identify missing phi nodes in a CFG.

14. Final Cheat Sheet

Item	Notes
Core definition	SSA operation that merges values at control-flow joins
Why it matters	Makes SSA possible for branches and loops
Most asked questions	What is phi, where placed, how lowered
Common comparisons	Phi vs assignment, phi vs ternary
One-line answer	Phi picks the correct SSA value based on which CFG edge entered the block.

11. Lowering High-Level Constructs

1. Overview

Definition: Lowering is the process of transforming high-level language constructs into simpler, lower-level IR constructs.

Examples:

- **for** loops become labels, condition checks, and jumps.
- **switch** statements become jump tables or chains of branches.
- Object method calls become function calls with explicit receiver arguments.
- Array access becomes address arithmetic plus load/store.

Why it matters:

- Machine code does not directly understand high-level constructs.
- Optimizers often work better on simpler IR.
- Lowering creates a bridge from source language features to target operations.
- It reveals the real runtime cost of high-level syntax.

Where it is used in real systems:

- C compilers lower loops, arrays, structs, and function calls.
- Java compilers lower enhanced **for** loops and lambdas.
- JavaScript engines lower dynamic operations into simpler IR with checks.
- Database engines lower SQL into relational algebra and physical plans.
- Backend frameworks lower high-level queries into SQL or execution plans.

Why interviewers ask about it: Lowering tests whether you understand what high-level constructs mean operationally. It is also useful for OA-style tracing and systems interviews.

2. Core Idea

Lowering removes syntactic convenience and exposes primitive operations.

Intuition: High-level code is nice for programmers; low-level IR is nice for compilers.

Real-world analogy: A restaurant order says "make a sandwich." The kitchen checklist lowers it into slice bread, add filling, toast, pack, serve.

Small example:

High-level:

```
for (i = 0; i < n; i++) {  
    sum += a[i];  
}
```

Lowered TAC-like IR:

```
i = 0  
L1:  
if i >= n goto L2  
t1 = i * 4  
t2 = a[t1]
```

```
sum = sum + t2
i = i + 1
goto L1
L2:
```

Step-by-step:

- 1. Initialization happens once.
- 2. Loop header checks condition.
- 3. False condition exits.
- 4. Body executes.
- 5. Update runs.
- 6. Jump returns to condition.

3. Important Subtopics

Subtopic	What It Means	Why It Matters	Example	Common Interview Angle
Loop lowering	Convert loops to labels and jumps	Exposes control flow	while, for	"Translate loop to TAC."
If lowering	Convert conditionals to branches	Builds CFG	true/false labels	"Translate if-else."
Short-circuit lowering	Preserve lazy boolean evaluation	Avoid wrong side effects/errors	a && b	"Does RHS always evaluate?"
Array lowering	Convert index to address arithmetic	Shows memory layout	base + i * width	"Translate a[i]."
Switch lowering	Branch chain or jump table	Performance trade-off	dense cases	"How switch compiled?"
Function-call lowering	Make call protocol explicit	Arguments, return value	param, call	"Translate call."
Object lowering	Convert methods/fields to offsets/calls	Runtime layout	obj.m()	"How OOP maps lower?"
Exception lowering	Add exceptional control paths	Accurate CFG	throw/catch edges	"Why exceptions complicate CFG?"

4. Real-World Example

Enhanced for loop in Java:

```
for (String s : list) {  
    print(s);  
}
```

is lowered roughly into iterator operations:

```
Iterator<String> it = list.iterator();  
while (it.hasNext()) {  
    String s = it.next();  
    print(s);  
}
```

The programmer sees a clean loop, but the compiler/runtime uses method calls and a normal loop structure.

5. Diagrams / Mental Models

If-Else Lowering

```
if (cond) S1 else S2
```

Lowered:

```
if cond goto Ltrue  
goto Lfalse  
Ltrue:  
    code for S1  
    goto Lend  
Lfalse:  
    code for S2  
Lend:
```

For Loop Lowering

```
for (init; cond; update) body
```

Lowered:

```
init  
Lcheck:  
    if not cond goto Lend  
    body  
    update  
    goto Lcheck  
Lend:
```

Array Access Lowering

```
a[i] for int array, width = 4

offset = i * 4
address = base(a) + offset
value = *address
```

6. Common Interview Questions

Question	Clear Answer	Expected Key Points	Common Mistakes
What is lowering?	Transforming high-level constructs into simpler lower-level IR.	Desugaring, explicit control/data ops	Saying it is optimization only
Lower a for loop.	Init, condition label, body, update, back jump, exit.	Correct order	Putting update before body
Lower an if-else.	Conditional jump to true/false labels, join label.	Branch and join	Missing end jump
Why lower constructs?	Target machines and lower IR do not directly support high-level syntax.	Simpler codegen/optimization	Saying only makes code longer
How lower array access?	Compute offset using index and element width, then load/store.	Base + index * width	Forgetting element size
How lower short-circuit AND?	If left is false, skip right.	Preserve semantics	Evaluating both sides always
How lower switch?	Chain of comparisons or jump table depending case density.	Trade-off	Assuming always if-else chain
How lower function calls?	Evaluate args, pass params, call function, handle return.	Order and convention	Ignoring side effects in args
Does lowering change meaning?	It should preserve semantics while changing representation.	Semantic equivalence	Saying compiler may change behavior
Is lowering same as optimization?	No. Lowering simplifies representation; optimization improves performance/size.	Different goals	Mixing terms

7. Deep-Dive Questions

Question	Answer
Why must short-circuit lowering preserve evaluation order?	Because the right side may have side effects or may be unsafe to evaluate, such as <code>p != NULL && p->x > 0</code> .
How can switch lowering choose between branch chain and jump table?	Dense integer cases are good for jump tables. Sparse cases may be better as comparison chains or binary search.
How are objects lowered?	Field access becomes offset-based memory access, and method calls may become function calls with an explicit receiver pointer and virtual dispatch if needed.
Why can lowering too early hurt optimization?	Some high-level information may be lost. For example, array bounds, object structure, or loop intent may help early optimizations.
How do exceptions affect lowering?	Operations that can throw need edges to handler blocks, so simple straight-line code may gain hidden control-flow paths.

8. Comparison Tables

Desugaring vs Lowering vs Optimization

Feature	Desugaring	Lowering	Optimization
Main goal	Replace syntax sugar	Move to simpler IR level	Improve performance/size
Meaning preserved	Yes	Yes	Yes, under language rules
Example	enhanced for to iterator loop	loop to labels/gotos	remove redundant computation
Level	Source/high-level IR	IR pipeline	Any IR level

Branch Chain vs Jump Table for Switch

Feature	Branch Chain	Jump Table
Best for	Sparse cases	Dense integer cases
Time	O(number of cases)	O(1) dispatch after bounds check
Space	Low	Higher
Example	cases 1, 100, 10000	cases 1, 2, 3, 4

9. Common Mistakes

- Treating lowering as optional decoration.
- Forgetting to preserve side-effect order.

- Evaluating both sides of short-circuit expressions.
- Forgetting loop update position.
- Forgetting array element width.
- Thinking `switch` is always lowered one way.
- Confusing lowering with optimization.

10. Edge Cases / Special Cases

- `continue` in a `for` loop jumps to update, not directly to condition.
- `break` jumps to loop/switch exit.
- Short-circuit boolean lowering must avoid evaluating skipped expressions.
- Function argument evaluation order depends on language rules.
- Multidimensional array lowering depends on row-major vs column-major layout.
- Virtual method calls may require vtable lookup.
- Exceptions add non-obvious control-flow edges.

11. How to Explain in Interview

"Lowering converts high-level constructs into simpler IR while preserving meaning. For example, a `for` loop becomes initialization, a condition label, conditional jump, body, update, and back jump. This makes optimization and machine-code generation easier."

12. Quick Revision Notes

- Lowering = high-level construct to lower-level IR.
- It preserves semantics.
- Loops become labels and jumps.
- Arrays become address arithmetic.
- Short-circuit logic becomes branches.
- Switch may become branch chain or jump table.
- Interview trap: lowering is not the same as optimization.

13. Practice Tasks

1. Lower a `for` loop into TAC.
2. Lower `if (a && b)` preserving short-circuit behavior.
3. Lower `a[i][j]` for row-major layout.
4. Lower a `switch` with dense cases into a jump table idea.
5. Explain how a method call `obj.f(x)` can become a lower-level function call.

14. Final Cheat Sheet

Item	Notes
Core definition	Converting high-level constructs into simpler lower-level IR
Why it matters	Makes control flow, memory access, and calls explicit

Item	Notes
Most asked questions	Lower for/while/if/switch/array access
Common comparisons	Desugaring vs lowering vs optimization, branch chain vs jump table
One-line answer	Lowering rewrites programmer-friendly constructs into compiler-friendly primitive operations.

Final Compact Cheat Sheet: Intermediate Code Generation

Topic	Core Definition	Must-Remember Interview Point
Intermediate Representation	Compiler internal program form	Enables optimization and target independence
Abstract Syntax Tree	Simplified tree of source structure	AST is not the same as parse tree
Three-Address Code	Simple IR with at most two operands and one result	Complex expressions become temporaries
Quadruples	TAC stored as (op, arg1, arg2, result)	Explicit result makes code movement easier
Triples	TAC stored as (op, arg1, arg2)	Result is instruction position
Control-Flow Graph	Graph of possible execution paths	Nodes are usually basic blocks
Basic Blocks	Maximal straight-line code sequences	Leaders define block boundaries
SSA	Each variable version assigned once	Phi functions merge versions
DAG Expressions	Shared graph for expression dependencies	Helps common subexpression elimination
Phi Functions	SSA merge at CFG joins	Selects value based on predecessor edge
Lowering	Converts high-level constructs to simpler IR	Must preserve semantics and evaluation order

Most Asked Interview Tasks

1. Convert an arithmetic expression into TAC.
2. Represent TAC as quadruples and triples.
3. Identify leaders and form basic blocks.
4. Draw a CFG for if-else and loops.
5. Convert simple code into SSA form.

6. Add phi functions at branch joins.
7. Draw a DAG for repeated expressions.
8. Lower `for`, `while`, `if`, `switch`, and array access.
9. Compare AST, TAC, CFG, DAG, and SSA.
10. Explain why IR is needed instead of direct machine-code generation.

One-Line Master Interview Answer

"Intermediate code generation converts a semantically checked program into compiler-friendly IR such as AST, TAC, CFG, SSA, or DAG forms so the compiler can analyze, optimize, and finally generate efficient target code."